A Unified Approach to Interpreting Model Predictions

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Motivation:

- Complex models such as deep neural networks and ensemble methods achieve high accuracy.
- t is difficult to understand **why a model makes a particular prediction**.
- In many applications (healthcare, finance, security), users need explanations to trust and verify model decisions.

Goal:

- Develop a method to explain the prediction of any machine learning model.
- Identify the contribution of each feature to the final prediction.

Accuracy vs Interpretability

Model	Accuracy	Interpretability
Linear Regression	Medium	High
Decision Trees	Medium	Medium
Deep Neural Networks	High	Low

Complex models increase accuracy but reduce interpretability.

Problem Statement

Given a model:

$$f(x)$$

Prediction example:

$$f(x) = 300000$$

We want to answer:

Why did the model predict this value?

Which features contributed to the prediction?

Local vs Global Explanations

Global Explanation

- Explains overall model behavior

Local Explanation

- Explains a single prediction

This paper focuses on **local explanations**.

Explanation Model

Original model:

$$f(x)$$

Explanation model:

$$g(z')$$

Goal:

$$g(z') \approx f(x)$$

Explanation model should be:

- Simple
- Interpretable

Simplified Inputs

Explanation models use simplified inputs:

$$z' \in \{0, 1\}^M$$

- $1 =$ feature present
- $0 =$ feature missing

Mapping function:

$$x = h_x(z')$$

Transforms simplified input back to original input.

Additive Feature Attribution Model

Definition:

$$g(z') = \phi_0 + \sum_{i=1}^{M} \phi_i z'_i$$

Where

- ϕ_0 = base value
- ϕ_i = contribution of feature i
- z'_i = indicator variable

Prediction = Base value + Feature contributions.

Example Explanation

Prediction = \$300K

Feature	Contribution
Base Value	200K
Location	+50K
Size	+40K
Age	+10K
Total	300K

Explanation decomposes prediction into feature effects.

Explanation Methods

Explanation methods of same additive class:

- LIME
- DeepLIFT
- Layer-wise Relevance Propagation
- Shapley Regression Values
- Shapley Sampling Values

All follow additive feature attribution models.

Method 1: LIME

Local Interpretable Model-Agnostic Explanations.
Steps:

- Generate nearby samples
- Evaluate model predictions
- Fit a local linear model
- Use coefficients as feature importance

Optimization:

$$\xi = \arg\min_{g \in G} L(f, g, \pi_x) + \Omega(g)$$

- L = loss function
- π = locality kernel
- Ω = complexity penalty

LIME Drawbacks

- Uses heuristic weighting
- May violate consistency
- Feature importance may change unpredictably

DeepLIFT I

Working Idea

- Compare model output with a reference input.
- Measure contribution of each feature to the change in prediction.
- Propagate contributions backward through the network.

Output Difference

$$\Delta o = f(x) - f(r)$$

- x : input instance
- r : reference input
- $f(x)$: prediction for input
- $f(r)$: prediction for reference

DeepLIFT II

Feature Difference

$$\Delta x_i = x_i - r_i$$

- x_i : value of feature i
- r_i : reference value of feature i

Contribution

$$C_{\Delta x_i \Delta o}$$

- Contribution of feature i to output change

DeepLIFT III

Property

$$\sum_i C_{\Delta x_i \Delta o} = \Delta o$$

Drawbacks

- Depends on reference input
- Model-specific (mainly neural networks)
- Uses heuristic propagation rules
- Does not guarantee consistency

Layer-wise Relevance Propagation

Idea:

- Backpropagate prediction through network layers
- Assign relevance to neurons

Drawback:

- Depends heavily on network architecture
- Uses heuristic rules

Properties of a Good Explanation Model

- **Local Accuracy**
  The explanation model should match the original model prediction.
- **Missingness**
  Features that are missing should have zero contribution.
- **Consistency**
  If a feature has a larger impact on the model output, its attribution should not decrease.

Shapley Value Formula

Shapley value for feature i

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)]$$

Notation

- ϕ_i : Contribution (importance) of feature i
- F : Set of all features
- S : Subset of features not containing feature i
- $|S|$: Number of features in subset S
- $|F|$: Total number of features
- $f(S)$: Model prediction using feature subset S
- $f(S \cup \{i\})$: Prediction when feature i is added to subset S

Measures average marginal contribution of feature.

Desirable Properties

1. **Local Accuracy**

$$f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i$$

2. **Missingness**

$$z_i' = 0 \Rightarrow \phi_i = 0$$

3. **Consistency**

If feature influence increases, its attribution should not decrease.

SHAP Values I

SHAP values explain predictions using:

- Additive feature attribution model
- Shapley values from game theory

Prediction explanation:

$$f(x) = E[f(x)] + \sum \phi_i$$

Where $E[f(x)]$ is the expected model output.

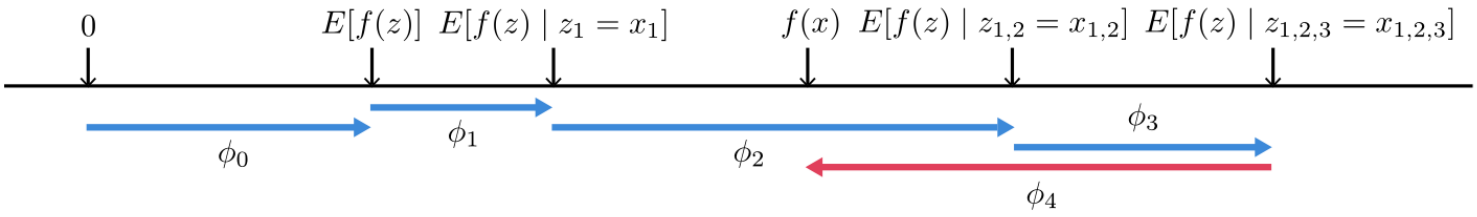

Figure: SHAP (SHapley Additive exPlanation) values attribute to each feature the change in the expected model prediction when conditioning on that feature. They

SHAP Values II

explain how to get from the base value E[f(z)] that would be predicted if we did not know any features to the current output f(x). This diagram shows a single ordering. When the model is non-linear or the input features are not independent, however, the order in which features are added to the expectation matters, and the SHAP values arise from averaging the i values across all possible orderings.

Kernel SHAP

Kernel SHAP estimates Shapley values using weighted regression.
Kernel function:

$$\pi_{x'}(z') = \frac{(M-1)}{\binom{M}{|z'|}|z'|(M-|z'|)}$$

Used for model-agnostic explanation.

Kernel SHAP Algorithm I

Input:

- Model f
- Instance x
- Background dataset Z

Steps

1. Generate binary feature vectors $z' \in \{0,1\}^M$ representing feature subsets.
2. Map each z' to the original feature space using the mapping function

$$x = h_x(z')$$

3. Evaluate the model output for each sample

$$y = f(h_x(z'))$$

Kernel SHAP Algorithm II

4. Assign weights to each sample using the Shapley kernel

$$\pi(z') = \frac{(M-1)}{\binom{M}{|z'|}\,|z'|(M-|z'|)}$$

5. Fit a weighted linear regression model

$$g(z') = \phi_0 + \sum_{i=1}^{M} \phi_i z'_i$$

6. Estimate SHAP values ϕ_i from the regression solution.

Output:

- SHAP values ϕ_i representing the contribution of each feature.

Kernel SHAP Algorithm III

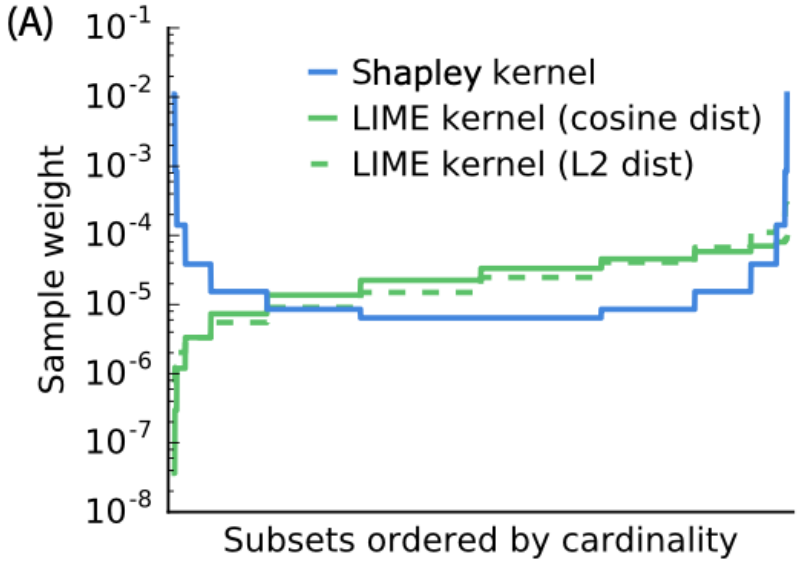

Figure: The Shapley kernel weighting is symmetric when all possible zvectors are ordered by cardinality there are 2^15 vectors in this example. This is distinctly different from previous heuristically chosen kernels.

Model-Specific Approximations: Linear SHAP

Model

$$f(x) = \sum_{j=1}^{M} w_j x_j + b$$

SHAP Values

$$\phi_0 = b$$

$$\phi_i = w_i(x_i - E[x_i])$$

Notation

- w_i : model weight for feature i
- x_i : value of feature i
- $E[x_i]$: expected value of feature i
- ϕ_i : SHAP value of feature i

Idea

Shapley Random Sampling

Idea

- Approximate Shapley values using random permutations of features.
- Avoid evaluating all 2^M feature subsets.

Steps

1. Generate random permutations of the feature set F.
2. For each permutation, add features sequentially.
3. Compute marginal contribution of feature i:

$$f(S \cup \{i\}) - f(S)$$

4. Repeat for many permutations.
5. Average the contributions to estimate ϕ_i.

Shapley Value Estimate

$$\phi_i = \frac{1}{K} \sum_{k=1}^{K} [f(S_k \cup \{i\}) - f(S_k)]$$

Low-order SHAP

Idea

- Approximate SHAP values using low-order feature interactions
- Avoid computing all 2^M subsets

Shapley Value

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)]$$

Approximation

- Consider only subsets with small size
- Ignore higher-order interactions

Benefit

- Reduced computational complexity

Max SHAP

Model

$$f(x) = \max(x_1, x_2, \ldots, x_M)$$

Idea

- Output determined by maximum feature value
- SHAP values computed using feature ordering

Contribution

$$\phi_i = E[\max(x_1, \ldots, x_M) \mid i \text{ enters}] - E[\max(x_1, \ldots, x_M) \mid i \text{ absent}]$$

Benefit

- Efficient computation for max operations
- Reduces complexity from exponential to polynomial

Deep SHAP (DeepLIFT + SHAP) I

Idea

- Combines DeepLIFT propagation rules with Shapley values
- Approximates SHAP values for deep neural networks

Deep SHAP multiplier

$$m_{x_j f_3} = \frac{\phi_i(f_3, x)}{x_j - E[x_j]}$$

$$\forall j \in \{1, 2\} \quad m_{y_i f_j} = \frac{\phi_i(f_j, y)}{y_i - E[y_i]}$$

$$m_{y_i f_3} = \sum_{j=1}^{2} m_{y_i f_j} \, m_{x_j f_3}$$

$$\phi_i(f_3, y) \approx m_{y_i f_3} \, (y_i - E[y_i])$$

Deep SHAP (DeepLIFT + SHAP) II

Notation

- x_j : input feature j
- y_i : hidden neuron value
- f_j : intermediate network component
- f_3 : final output of the network
- $\phi_i(\cdot)$: SHAP value contribution of feature i
- $E[x_j]$: expected (baseline) value of feature x_j
- $E[y_i]$: expected activation of neuron y_i

Deep SHAP (DeepLIFT + SHAP) III

Algorithm

1. Select background dataset.
2. Compute reference activations.
3. Apply DeepLIFT contribution rules.
4. Propagate contributions through network layers.
5. Average contributions across background samples.

Output

- SHAP values ϕ_i for each input feature.

Deep SHAP (DeepLIFT + SHAP) IV

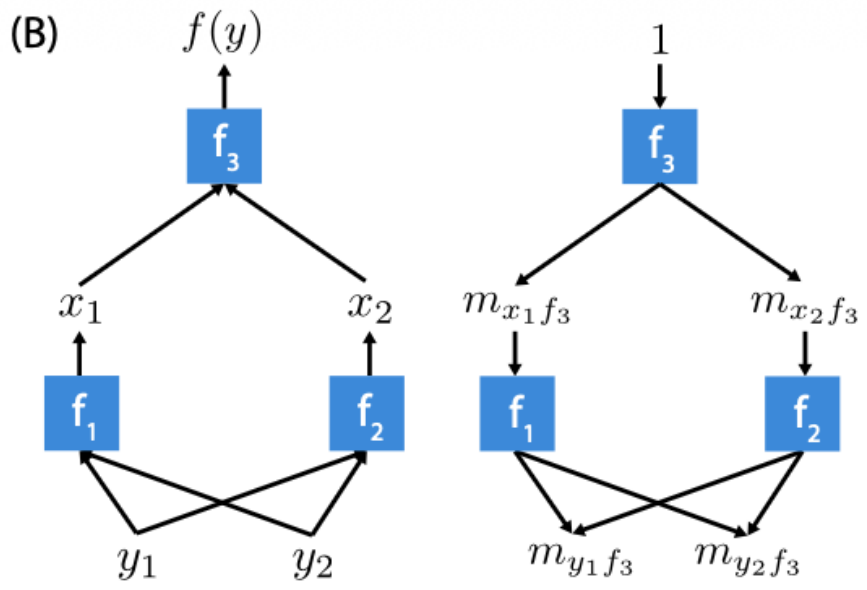

Figure: Compositional models such as deep neural networks are comprised of many simple components. Given analytic solutions for the Shapley values of the components, fast approximations for the full model can be made using DeepLIFT's style of back-propagation.

Comparisions I

Paper compares:
SHAP LIME Shapley sampling
Results show: SHAP explanations are more consistent

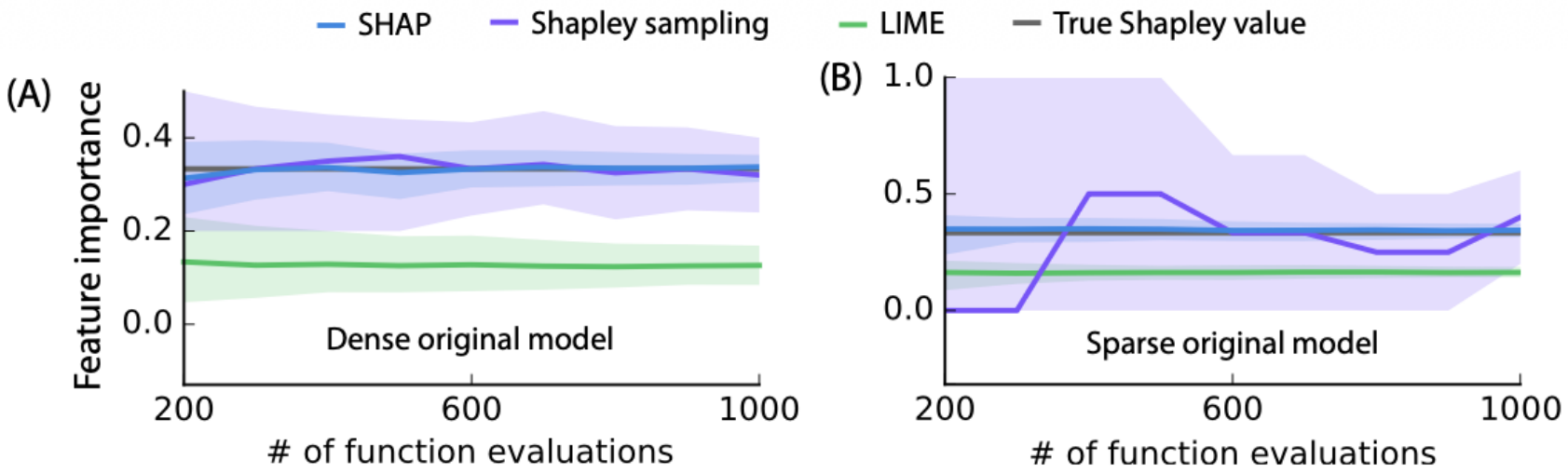

Figure: Comparison of three additive feature attribution methods: Kernel SHAP (using a debiased lasso), Shapley sampling values, and LIME (using the open source implementation). Feature importance estimates are shown for one feature in two models as the number of evaluations of the original model function

Comparisions II

increases. The 10th and 90th percentiles are shown for 200 replicate estimates at each sample size. (A) A decision tree model using all 10 input features is explained for a single input. (B) A decision tree using only 3 of 100 input features is explained for a single input.

Experimental Results I

Computational Efficiency

- Kernel SHAP estimates SHAP values using fewer model evaluations.
- Faster convergence compared to Shapley Sampling.
- Provides more accurate feature importance estimates than LIME.

Consistency with Human Intuition

- Human studies performed on feature attribution tasks.
- SHAP explanations better matched human reasoning.
- Demonstrates interpretability and fairness of SHAP values.

Explaining Class Differences (MNIST)

- SHAP applied to convolutional neural network for digit classification.
- Identifies pixels contributing to the prediction.
- Removing SHAP-important features significantly changes model output.

Experimental Results II

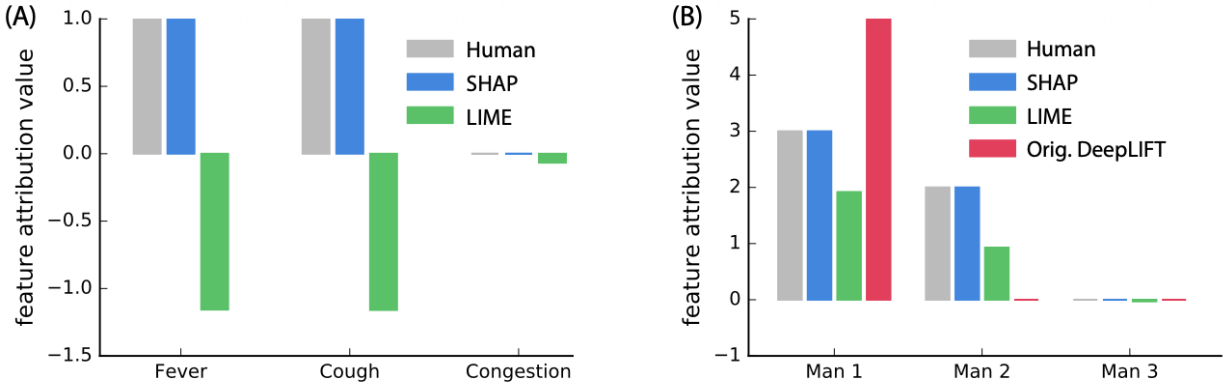

Figure: Human feature impact estimates are shown as the most common explanation given among 30 (A) and 52 (B) random individuals, respectively. (A) Feature attributions for a model output value (sickness score) of 2. The model output is 2 when fever and cough are both present, 5 when only one of fever or cough is present, and 0 otherwise. (B) Attributions of profit among three men, given according to the maximum number of questions any man got right. The first man got 5 questions right, the second 4 questions, and the third got none right, so the profit is 5.

Experimental Results III

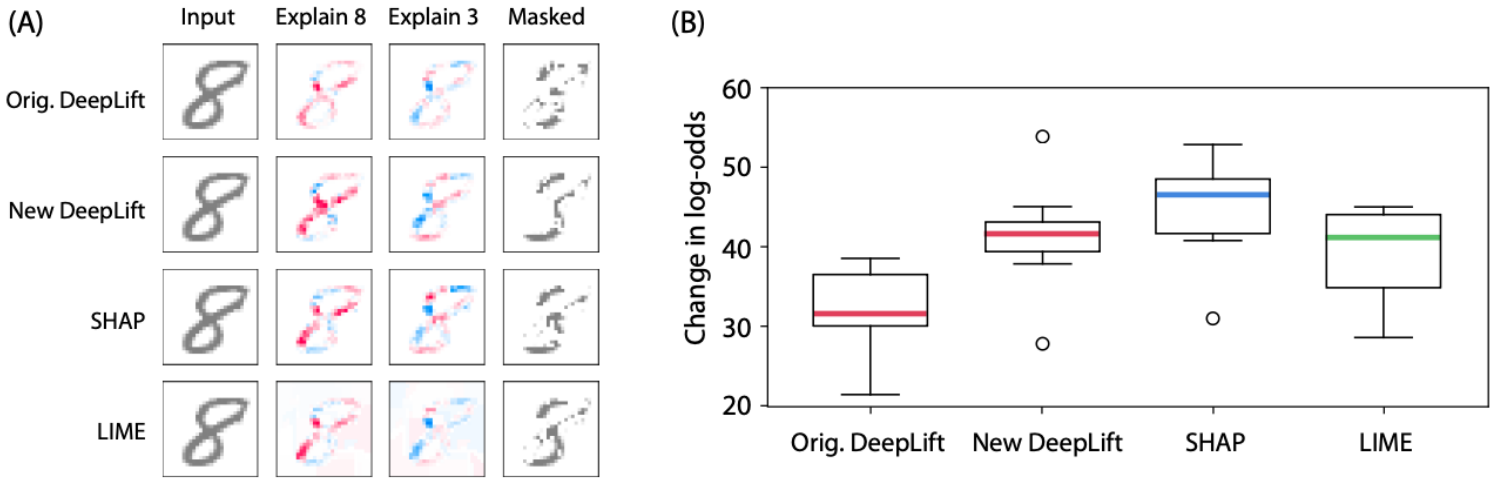

Figure: Explaining the output of a convolutional network trained on the MNIST digit dataset. Orig. DeepLIFT has no explicit Shapley approximations, while New DeepLIFT seeks to better approximate Shapley values. (A) Red areas increase the probability of that class, and blue areas decrease the probability. Masked removes pixels in order to go from 8 to 3. (B) The change in log odds when masking over 20 random images supports the use of better estimates of SHAP values.

Advantages of SHAP

- Theoretically grounded
- Consistent explanations
- Model-agnostic
- Unified framework

Conclusion

Main contributions of the paper:

- Unified framework for explanation methods
- Proved Shapley values are unique solution
- Introduced SHAP algorithms

Key Idea:
SHAP explains predictions by fairly distributing model output among input features.

Thank You.